

[1.3.55]

$$A \xrightarrow{f} B \xrightarrow{g} C$$

- (1) $g \circ f$ - injectivā dar g - mē injectivā
 (2) $g \circ f$ - surjectivā dar f - mē surjectivā
 (3) $g \circ f$ - bijectivā dar g - mē injectivā un f - mē surjectivā

$$g \circ f \text{ - inj } \Leftrightarrow f \text{ - inj}$$

MY TAKES

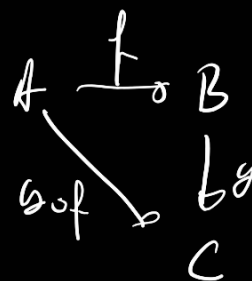
(1) $A = \{a_1\}$

$B = \{b_1, b_2\}$

$C = \{c_1\}$

$$(g \circ f)(x) = g(f(x))$$

$$g \circ f \text{ - inj } \Leftrightarrow \forall$$



ar $(g \circ f)(a_1) = c_1$

$$f(a_1) = b_1 \Rightarrow f \text{ - injectivā}$$

$$\left. \begin{array}{l} g(b_1) = c_1 \\ g(b_2) = c_1 \end{array} \right\} \Rightarrow g \text{ - mē injectivā}$$

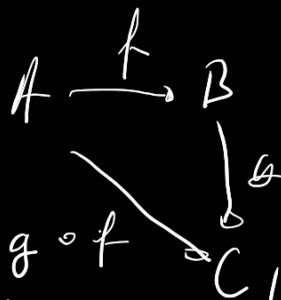
(2) $g \circ f$ - surjectivā $\Leftrightarrow g$ - surjectivā

$A = \{a_1\}$ $C = \{c_1\}$

$B = \{b_1, b_2\}$

$f(a_1) = b_1 \Rightarrow \exists b \in B$ ar $\forall a \in A, f(a) = b \Rightarrow f$ - mē surjectivā

$g(b_1) = c_1, g(b_2) = c_1 \Rightarrow \forall c \in C \exists b \in B$ ar $g(b) = c \Rightarrow g$ - surj.



(3) $g \circ f$ - bijectivă $\Leftrightarrow$

$g \circ f$ - injectivă, dacă f - inj
 $g \circ f$ - surjectivă, dacă g - surj

$$\nexists a_1, a_2 =$$

$$f \text{ - inj} \quad \forall b \in B \quad \exists! a \in A \text{ aî. } f(a) = b$$

$$A = \{a_1\} \quad f(a_1) = b_1 \quad \checkmark \quad a_1 \mapsto b_1$$

$$B = \{b_1, b_2\} \quad C = \{c_1\}$$

$$\left(\nexists a_1, a_2 \quad a_1 \neq a_2 \in A \quad f(a_1) \neq f(a_2) \rightarrow a_1 \neq a_2 \right.$$

$$g(b_1) = c_1 \quad \Rightarrow \quad \forall c \in C \quad \exists b \in B \text{ aî. } g(b) = c \quad \xrightarrow{g \text{ surj}}$$

$$g(b_2) = c_1 \quad \Rightarrow g \text{ - nu e inj.}$$

1.3.42

$$f: A \rightarrow B = ? \text{ aî.}$$

(1) f - inj dar nu are o inversă la stînga
 o funcție are inversă la stînga $\Leftrightarrow$ este injectivă

$\hookrightarrow \Rightarrow$ NU SE PATE ???

(2) f - are exact o inversă la stînga, dar nu e bijectivă
 o f are exact o singură inversă la stînga $\Leftrightarrow$ e bijectivă

$\hookrightarrow \Rightarrow$ —, — ???

(3) f are exact două inverse la stânga

1.3.43

$g: B \rightarrow A$ aî.

(1) g - exact două inverse la dreapta

(2) g are o infinitate de inverse la dreapta

+ Să se arate că g are exact o inversă la dr. dacă e
bijectivă

